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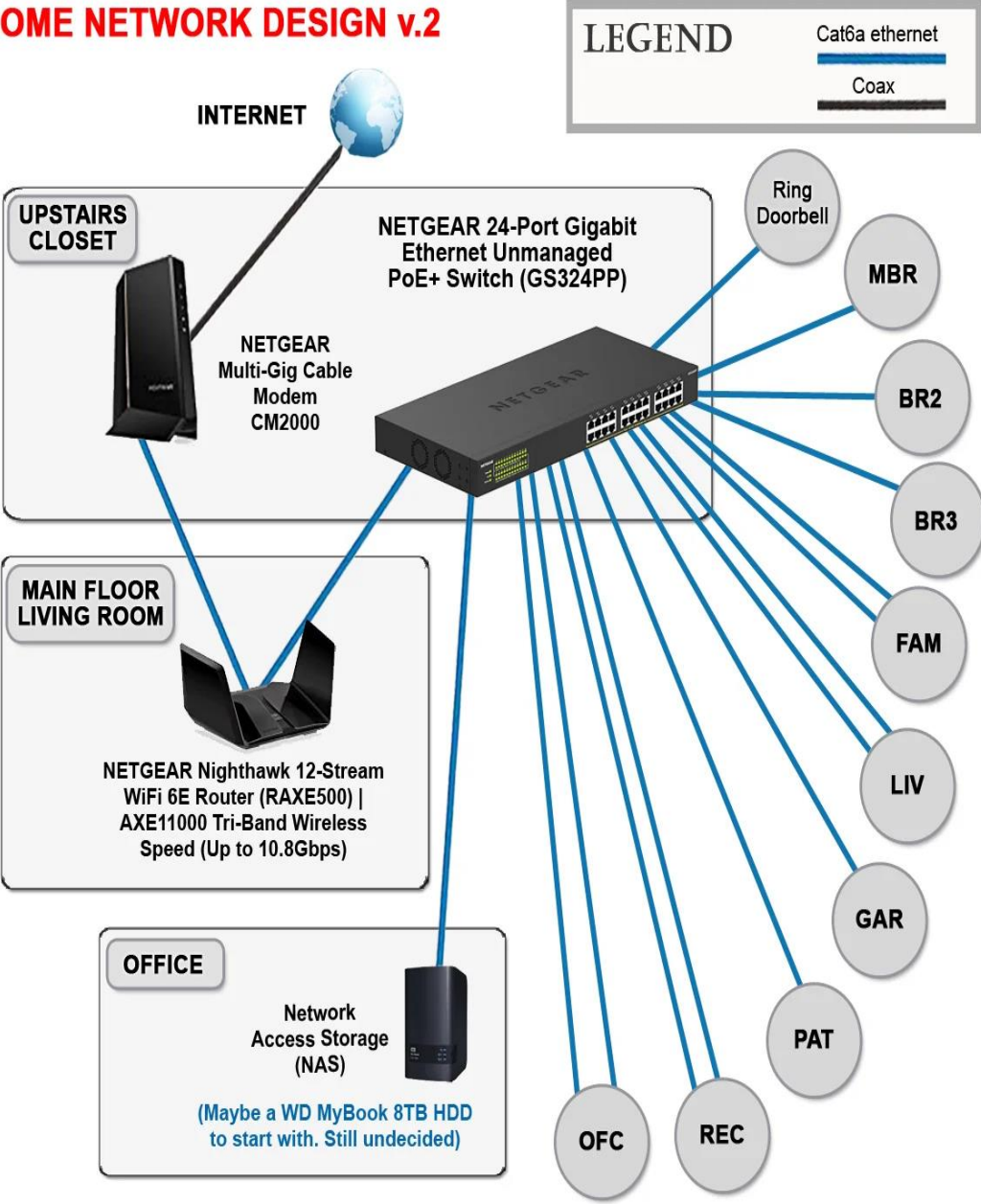
REGISTRAION NO. : 2141016019

CA-NETWORK (CA-N)

ASSIGNMNET - 1

Assignment 1: Draw your Home Network Topology and explain it.

HOME NETWORK DESIGN v.2



Home Network Topology Explained

1 Internet Connection

- ISP (Internet Service Provider):
Provides internet access to your home.
- Modem: Connects to the ISP's infrastructure (via cable, DSL, or fiber) and translates the signal for your home network.

2 Router

- Function: Distributes the internet connection from the modem to multiple devices within your home.
- Features:
 - DHCP Server: Assigns IP addresses to devices on the network.
 - Firewall: Provides basic security by controlling incoming and outgoing traffic.
 - NAT (Network Address Translation): Allows multiple devices to share a single public IP address.

3 Wireless Access Point (WAP)

- Often integrated into the router
- Provides wifi connectivity to wireless devices like smartphones, tablets, and laptops.

4 Switch

- Function: Expands the number of available ethernet ports, allowing more wired devices to connect to the network.
- Use Case: Ideal for connecting devices that require stable and high-speed connections, such as desktops PCs, gaming consoles, and smart TVs.

5 Connected Devices

- Wired Devices:
 - Desktops PCs
 - Gaming Consoles
 - Smart TVs
 - Network Attached Storage (NAS)
- Wireless Devices:
 - Laptops
 - Smartphones
 - Tablets
 - Smart Home Devices (e.g., smart thermostats, security cameras)

Tips for an Efficient Home Network.

- **Placement of Router:** Position your router centrally to ensure optimal Wi-Fi coverage throughout your home.
- **Use of switches:** If you have multiple wired devices, consider using a switch to expand your network without overloading the router.
- **Network Segmentation:** For enhanced security, segment your network to separate IoT devices from personal computers.
- **Regular Updates:** Keep your router's firmware up to date to benefit from security patches and performance improvements.